

Vitamin D Supplementation and the Incidence of Fractures in the Elderly Healthy Population: A Meta-analysis of Randomized Controlled Trials



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ABSTRACT

INTRODUCTION: Although a well-established component of bone metabolism, the efficacy and safety of vitamin D supplementation for the prevention of fractures in elderly healthy individuals is still unclear.

PURPOSE: To perform a meta-analysis comparing vitamin D supplementation with placebo and its contributions on fracture incidence.

METHODS: This meta-analysis was registered in the International Prospective Register of Systematic Reviews (PROSPERO), under protocol CRD42023484979. We systematically searched PubMed, Embase, and Cochrane Central databases from inception to November 2023 for randomized controlled trials (RCTs) comparing vitamin D supplementation versus placebo in individuals with 60 years of age or more and without bone related medical conditions such as cancer and osteoporosis.

RESULTS: Seven RCTs with 71,899 patients were included, of whom 36,822 (51.2%) were women. There was no significant difference in total fracture incidence (RR 1.03; 95% CI 0.93–1.14; $p=0.56$; $I^2=58\%$) between groups or subgroups. However, women had an increased risk for hip fractures (164 vs. 121 events; RR 1.34; 95% CI 1.06–1.70; $p=0.01$; $I^2=0\%$). There was no significant difference in non-vertebral fractures, osteoporotic fractures development, or falls (RR 1.02; 95% CI 0.94–1.12; $p=0.6$; $I^2=47\%$; RR 0.97; 95% CI 0.87–1.08; $p=0.63$; $I^2=0\%$; RR 1.01; 95% CI 0.97–1.04; $p=0.66$; $I^2=55\%$, respectively).

CONCLUSION: Vitamin D supplementation does not reduce the total fracture development rate in the elderly healthy population, and it may increase the incidence of hip fractures among elderly healthy women. This finding suggests refraining from prescribing high intermittent doses of vitamin D, without calcium, to individuals aged 60 or older with unknown vitamin D serum concentration or osteoporosis status and inadequate calcium intake.

KEY WORDS: vitamin D; supplements; fracture; hip fractures; elderly

J Gen Intern Med 39(14):2829–36

DOI: 10.1007/s11606-024-08933-1

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INTRODUCTION

Vitamin D has a well-established role in bone health, as low concentrations are associated with impaired calcium absorption and a compensatory rise in parathyroid hormone (PTH), which results in excessive bone resorption.^{1–4} Individuals with osteoporosis present low bone density, deterioration of bone tissue, disruptions of bone microarchitecture, and an increased risk for fractures.⁴ The association between hypovitaminosis D and osteoporosis is well-studied as a risk factor for developing the condition.⁴

Routine assessments of 25-hydroxyvitamin D (25(OH)D) levels are conducted in patients with osteoporosis, with the standard recommendation of vitamin D and calcium supplementation as part of the treatment of osteoporosis. The recommendations of the Endocrine Society for prevention disease differ depending on age. In healthy individuals aged 50 to 74 years, it suggests against routine 25(OH)D testing and vitamin D supplementation. However, for the population aged 75 years and older, it suggests empiric vitamin D supplementation with 600 IU daily for individuals aged 50 to 70 years, and 800 IU daily for those older than 70 years, for the potential decreased risk of mortality.^{5,6} However, the impact of vitamin D supplementation alone in primary prevention of fracture in elderly individuals without osteoporosis and hypovitaminosis D is a subject of controversy. For instance, two previous meta-analyses observed positive outcomes with vitamin D supplements (≥ 800 IU daily) in preventing hip and non-vertebral fractures in individuals aged 65 years and older,⁷ as well as its combination with calcium demonstrated a reduction in overall fractures and hip fractures.⁸ However, their conclusions may be influenced by participants residing in institutional settings. Conversely, other published meta-analyses demonstrated a neutral impact of vitamin D supplementation and the incidence of fractures.^{9,10}

Received April 17, 2024

Accepted June 28, 2024

Published online July 12, 2024

Therefore, we sought to perform an updated systematic review and meta-analysis examining the impact of vitamin D supplementation alone, without calcium, compared to placebo on fracture outcomes in healthy participants aged 60 years or more. We hypothesize that such supplementation does not prevent fractures in this population.

METHODS

This systematic review and meta-analysis were performed and reported in accordance with the Cochrane Collaboration Handbook for Systematic Reviews of Interventions and the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) Statement Guidelines.^{11,12}

Eligibility Criteria

Inclusion in this meta-analysis was restricted to studies that met all the following eligibility criteria: (1) randomized controlled trial; (2) comparing vitamin D in an average daily dose of 800 IU or higher with placebo (since that is the minimum dosage recommended for the main guidelines for that population^{3,6}); (3) enrolling patients with more than 60 years old; (4) with follow-up with at least 12 months; and (5) included at least 500 participants, in order to minimize random error and publication bias. In addition, studies were included only if they reported fractures as the clinical outcomes of interest. We excluded studies that had (1) calcium supplementation on the intervention; (2) participants living in nursing homes; and (3) participants with medical conditions related with bone metabolism, such as cancer, osteoporosis, nephrolithiasis, sarcoidosis, or parathyroid disorders.

Search Strategy and Data Extraction

We systematically searched PubMed, Embase, and Cochrane Central databases from inception to November 2023 with the main following search terms: “Vitamin D,” “supplementation,” “elderly,” “fracture,” and “randomized trial.” Search strategies used in each databases are presented in the Appendix. The references of all included studies and previous meta-analyses were also searched manually for any additional studies. Study triage against eligibility criteria was performed by two authors (M.S. and R.D.). Disagreements were resolved through a consensus after discussing reasons for discrepancy.

Two authors (M.S. and V.D.) independently extracted the data following predefined search criteria and quality assessment. Disagreements were resolved through a consensus after discussing reasons for discrepancy. The prospective meta-analysis protocol was registered in the International

Prospective Register of Systematic Reviews (PROSPERO) on November 20th, 2023, under protocol CRD42023484979.

Endpoints and Subgroup Analyses

Outcomes included total fractures, hip fractures, osteoporotic fractures, non-vertebral fractures, vertebral fractures, and falls.

Prespecified subgroups analyses included data stratified by (1) gender and (2) equivalent dose of vitamin D (≥ 1500 IU/day and < 1500 IU/day).

Leave-one-out sensitivity analyses were performed in total fractures and hip fractures outcomes.

Sensitivity analyses were performed removing studies with a high risk of bias and removing the only trial that used daily vitamin D supplementation.

Quality Assessment

We evaluated the risk of bias for the included studies using version 2 of the Cochrane Risk of Bias assessment tool (RoB 2).¹² Each trial received a score of high, low, or unclear risk of bias in five domains: selection, performance, detection, attrition, and reporting biases. Two independent authors completed the risk of bias assessment (M.S. and R.D.). Disagreements were resolved through a consensus after discussing reasons for discrepancy.

Small-study effect was investigated by funnel-plot analysis of point estimates in relation to study weights.

Statistical Analysis

Statistical analyses were performed according to the Cochrane Collaboration Handbook for Systematic Reviews of Interventions¹² and in line with the PRISMA statement.¹¹ Pooled risk ratio (RR) and hazard ratio (HR) with associated 95% confidence intervals were used to compare treatment effects for binary endpoints with an intention-to-treat basis. Heterogeneity was evaluated with Cochran Q test and I^2 statistics; p values inferior to 0.10 and $I^2 > 25\%$ were considered significant for heterogeneity. We used DerSimonian and Laird random-effects model due to anticipated heterogeneity between studies in clinical practice. Review Manager 5.4 (Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark) was used for statistical analysis. R version 4.3 was used to perform funnel plots of study weights vs. point estimates and Egger's test.

RESULTS

Study Selection and Characteristics

As detailed in Supplemental Fig. 1, 2160 studies were identified after databases searches. After removal of duplicate articles and unrelated studies, 27 RCTs were fully reviewed against the inclusion and exclusion criteria. A total of seven studies^{13–19} with 71,899 participants were included, of whom

Table 1 Baseline Characteristics of Included Studies

Study	Dose of vitamin D	Follow-up, months	Patients, no	Age, years [†]	Female sex, no. (%)	BMI (kg/m ²) [‡]	Serum 25-hydroxyvitamin D (ng/ml) [‡]	Mean daily calcium intake (mg)
VITAL 2022	2000 IU/day	63	25871	67.1	13,085 (50.6)	28.1	30.7	NA
TIPS 2022	60,000 IU/month	55	5670	63.9	3005 (52.9)	25.8	NA	NA
VIDA 2017	100,000 IU/month	48	5108	68.8	2139 (37.7)	28.5	NA	NA
VITAL D 2010	500,000 IU/year	48	2256	76.3 [§]	2256 (100)	NA	19.9 [§]	976 mg
Smith 2007	300,000 IU/year	36	9440	79.5 [§]	5086 (53.8)	NA	NA	625 mg
D HEALTH 2023	60,000 IU/month	63.6	20,326	NA	9295 (45.7)	NA	NA	NA
Trivedi 2003	100,000 IU/4 month	60	2686	76.5 [†]	649 (24.1)	24.3	NA	742 mg

BMI body mass index, NA not available

[†]Mean

[§]Value estimated mean of the sample using the number of the sample, median, and first and third quartile of the sample from Luo et al. (2018)²⁰

36,822 (51.2%) were women. Population baseline characteristics are presented in Table 1.

All studies used oral supplements of vitamin D, but were heterogeneous with regard to vitamin D dosage. A total of 25,916 (78.2%) participants received a daily dose equivalent of vitamin D higher than 1500 IU. Six trials^{13–16, 18, 19} used monthly or annual doses of vitamin D, whereas one study prescribed a daily dose of 2000 IU.¹⁷

The mean calcium intake of participants from three trials^{13,18, 19} is provided in Table 1. One trial¹⁷ reported that 20% of participants used calcium supplements at a dosage of ≤ 1200 mg per day, while in another trial,¹⁶ 5% of participants used calcium supplements (dosage not specified).

Pooled Analysis of All Studies

The incidence of total fractures was not significantly different between participants who received vitamin D or placebo (RR 1.03; 95% CI 0.93–1.14; $p=0.56$; $I^2=58\%$; Fig. 1A)/HR 1.03; 95% CI 0.95–1.12; $p=0.44$; $I^2=28\%$; Fig. 1B). There was no difference in total fractures regarding vitamin D dosage analysis (<1500 IU/day RR 1.04; 95% CI 0.82–1.32; $p=0.76$; $I^2=77\%$; Fig. 1A/ ≥ 1500 IU/day RR 1.01; 95% CI 0.91–1.11; $p=0.56$; $I^2=58\%$; Fig. 1B). Similar results were found for both females (RR 1.06; 95% CI 0.91–1.22; $p=0.46$; $I^2=65\%$; Fig. 2A/HR 1.06; 95% CI 0.90–1.24; $p=0.45$; $I^2=57\%$; Fig. 2B) and males (RR 0.94; 95% CI 0.81–1.09; $p=0.42$; $I^2=43\%$; Fig. 2A/HR 0.93; 95% CI 0.74–1.16; $p=0.51$; $I^2=58\%$; Fig. 2B).

There was a significant 34% relative risk increase in hip fractures among women treated with vitamin D compared with placebo (164 vs. 121 events in 14,033 participants) (RR 1.34; 95% CI 1.06–1.70; $p=0.01$; $I^2=0\%$; Fig. 3A). The same result was found when time was evaluated (HR 1.42; 95% CI 1.11–1.81; $p=0.005$; $I^2=0\%$; Fig. 3B). In the subgroup of men, there was not a significantly difference in hip fractures for vitamin D versus placebo (RR 0.86; 95%

CI 0.65–1.13; $p=0.28$; $I^2=0\%$; Fig. 3A)/HR 0.87; 95% CI 0.65–1.18; $p=0.37$; $I^2=0\%$; Fig. 3B).

The incidence of non-vertebral fractures (RR 1.02; 95% CI 0.94–1.12; $p=0.6$; $I^2=47\%$; Supplemental Fig. 2A/HR 1.00; 95% CI 0.92–1.10; $p=0.95$; $I^2=36\%$; Supplemental Fig. 2B) and osteoporotic fractures were not significantly different between patients who received vitamin D or placebo (RR 0.97; 95% CI 0.87–1.08; $p=0.63$; $I^2=0\%$; Supplemental Fig. 3A/HR 1.00; 95% CI 0.89–1.12; $p=0.94$; $I^2=0\%$; Supplemental Fig. 3B). The results were similar in the subgroup analyses between women and men.

Falls were not significantly different between participants who received vitamin D or placebo (RR 1.01; 95% CI 0.97–1.04; $p=0.66$; $I^2=55\%$; Supplemental Fig. 4A/HR 1.02; 95% CI 0.93–1.12; $p=0.69$; $I^2=0.72$; Supplemental Fig. 4B).

Sensitivity Analyses

When we excluded the only trial that used a daily dose of vitamin D as an intervention¹⁷ (2000 IU/day), the result in hip fractures among women was statistically non-significant (Supplemental Fig. 5).

We performed a sensitivity analysis in the incidence of hip fractures removing the trials with high risk of bias,^{15,19} and the result did not change among women participants (RR: 1.13; 95% CI 0.87–1.47; $p=0.01$; $I^2=0\%$; Supplemental Fig. 6).

Leave-one-out sensitivity analyses on total fractures (Supplemental Fig. 7) and hip fractures (Supplemental Fig. 8) did not change the result of which outcome among total participants.

Risk of Bias Assessment

Supplemental Table 1 outlines the individual appraisal of each RCT included in the meta-analysis. Five^{13,14, 16, 17, 21} of them were deemed at low risk or some concerns of bias, while two studies^{15,19} were considered at high risk of bias, both in the measurement of outcomes category.

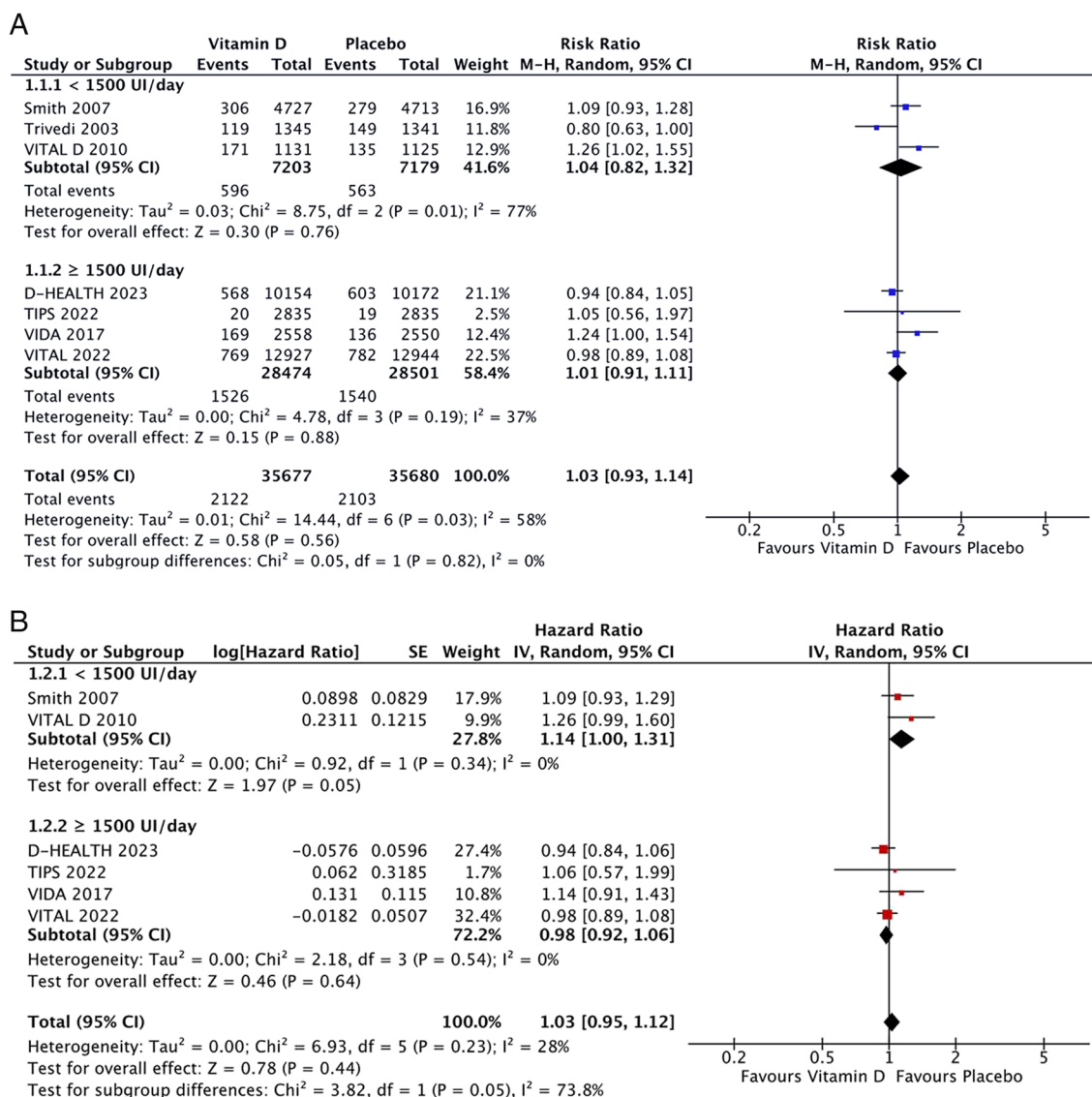


Figure 1 A The incidence of total fractures was similar in vitamin D and placebo group regarding vitamin D dosage ($p = 0.56$). **B** The incidence of total fractures was similar in vitamin D and placebo group regarding vitamin D dosage ($p = 0.44$).

Regarding publication bias, linear regression by Egger's test does not indicate funnel-plot (Supplemental Fig. 9) asymmetry on total fractures (Egger -0.045 ; 95% CI -2.14 – 2.05 ; $p = 0.96$).

DISCUSSION

In this systematic review and meta-analysis of seven RCTs involving a total of 71,899 participants, we investigated the efficacy of vitamin D supplementation compared with placebo for fracture prevention in elderly healthy individuals. The main findings were as follows: (1) vitamin D increased the relative risk of hip fractures by 34% among women; (2) there was no significant difference between vitamin D or placebo and the incidence of total fractures, non-vertebral fractures, osteoporotic fractures, and falls for this population.

To the best of our knowledge, our study stands out as the first to assess the consequences of vitamin D supplementation alone in a population without major diseases and only community-dwelling, and it is also the largest in terms of participant numbers ever conducted.

Previous meta-analyses have suggested that the benefits of vitamin D supplementation may be limited to individuals residing in institutional settings, as they are a population more prone to develop osteoporosis and fractures due to factors such as immobility, reduced sun exposure, and poor diet.^{7,8} In addition, several published meta-analyses have shown that calcium supplements in combination with vitamin D may have an important role in fracture prevention.^{9,22,23}

From a pathophysiology perspective, when calcium intake is insufficient, a transient hypocalcemia occurs,

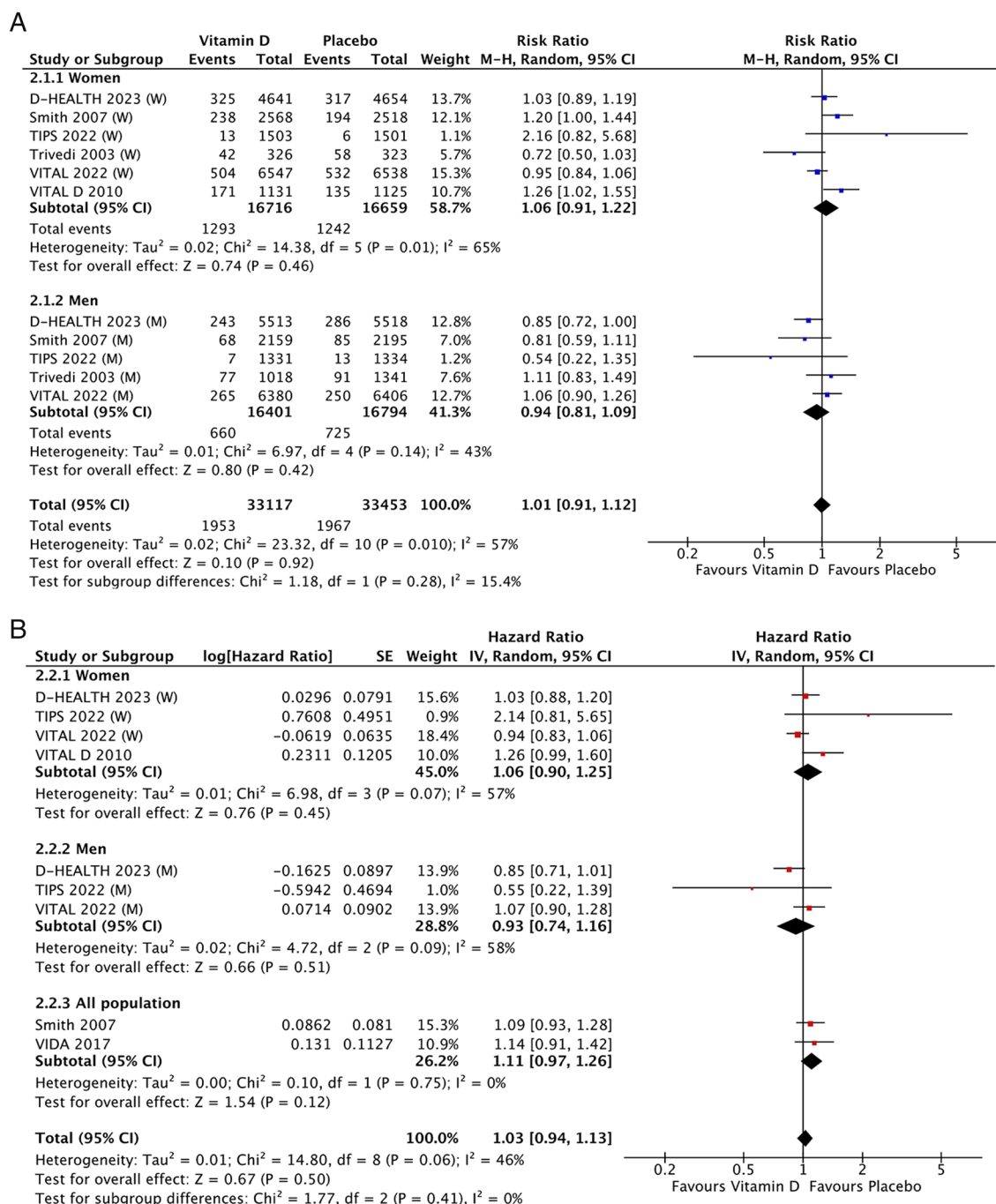


Figure 2 A The incidence of total fractures was similar in vitamin D and placebo group regarding subgroup analysis by gender ($p = 0.92$).
B The incidence of total fractures was similar in vitamin D and placebo group regarding subgroup analysis by gender ($p = 0.50$).

which induces an increased secretion of parathyroid hormone (PTH). This relative hyperparathyroidism provokes higher renal calcium reabsorption, as well as calcium mobilization from the bones, hampering bone mineralization.^{24,25} These factors may have contributed negatively to fracture risk, despite adequate vitamin D level supplementation.^{26–30}

Estrogen deficiency significantly impacts bone density reduction. The average annual rate of spinal bone loss measured by bone density scan is approximately 2.5% for the first

2 years after menopause, 1.8% in the next 2 years, and about 1% annually for subsequent years.³¹ Bone loss, even in non-osteoporotic women, may be associated with increased fracture rates. In an observational cohort study involving more than 200,000 post-menopausal women, 2259 experienced new fractures, but only 6.4% of these women were diagnosed with osteoporosis.³²

A previous meta-analysis reported a vitamin D dose-response relationship with a greater reduction in

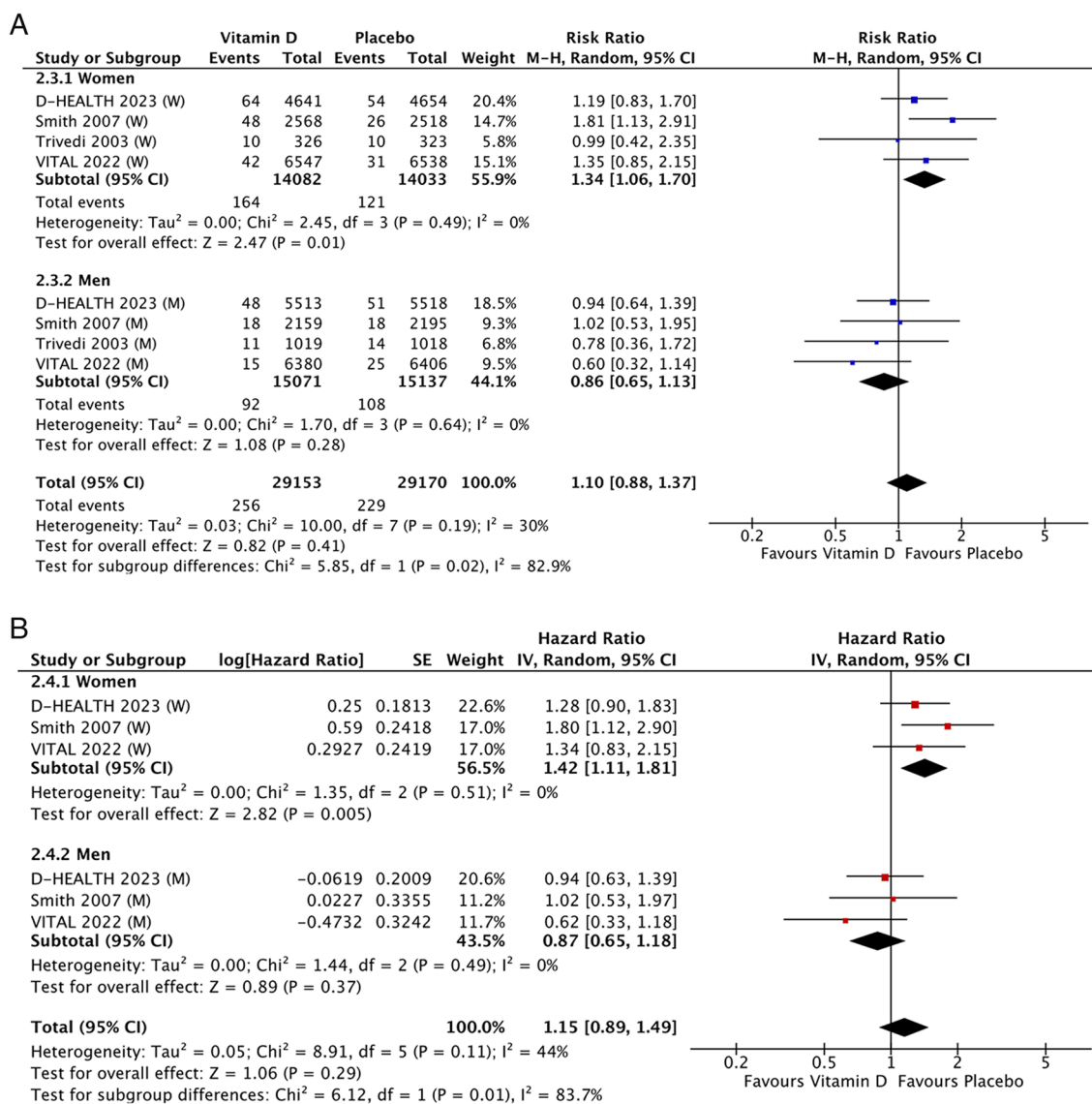


Figure 3 A The incidence of hip fractures in women were significantly higher in the vitamin D group in women ($p = 0.01$). The incidence of hip fractures in men and in general population were similar between groups ($p = 0.28$ and $p = 0.41$, respectively). B The incidence of hip fractures in women were significantly higher in the vitamin D group in women ($p = 0.005$). The incidence of hip fractures in men and in general population were similar between groups ($p = 0.37$ and $p = 0.29$, respectively).

fracture incidence with higher doses.⁷ However, this result may have been influenced by the fact that vitamin D was predominantly administered daily, since sensitivity analysis including a trial with high annual doses of vitamin D attenuated the benefit.⁷ In our meta-analysis, high intermittent doses of vitamin D were used in six out of the seven included trials and the mean prescribed vitamin D dosage across the trials was 1769 IU/day, more than twice the recommendation by the Institute of Medicine,⁶ which may justify the neutral or negative consequences in relation to fractures we have found.

A published RCT compared weekly versus monthly vitamin D supplements intake and concluded that the group who received the intervention monthly had the highest adherence and was preferred by the participants.³³ Moreover, recent

RCTs have explored vitamin D dosing interval and medication adherence, with Sanders et al. (2010)¹³ revealing potential adverse effects, such as increased incidence of falls and fractures, associated with very high intermittent doses (500,000 IU per year). In our sensitivity analysis focusing only trials with high and intermittent doses, we removed the VITAL 2022¹⁷ which reported neutral results regarding hip fractures in women when comparing vitamin D to placebo. After removing this trial, our meta-analysis continued to indicate an increased risk of hip fractures among women, although this finding did not reach statistical significance, likely due to reduced statistical power resulting from the exclusion of one of the largest trials.

This study has limitations. Firstly, it is important to highlight that two of the included studies have a high risk of bias

in outcomes measurement, potentially impacting obtained data, although their removal did not change the results. Secondly, there is a relative small number of RCTs studying senile and healthy individuals. Furthermore, there is a significant variability in vitamin D doses and intervals across the included trials, and other risk factors such as sarcopenia, sedentary lifestyle, and lack of strength training, as well as undiagnosed renal dysfunction or malabsorption, were not evaluated in all individual trials included in this meta-analysis. Additionally, more studies focusing on optimal dosing regimens and longer follow-up periods are necessary to comprehensively evaluate outcomes.

CONCLUSION

Vitamin D supplementation, especially in high and intermittent doses, and without adequate calcium intake, is not associated with fracture development prevention and might increase the incidence of hip fracture in older healthy women. This study provides evidence supporting the recommendation to refrain from prescribing high intermittent doses of vitamin D supplements, without calcium, to individuals aged 60 or older with unknown osteoporosis and inadequate calcium intake.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11606-024-08933-1>.

Acknowledgements: Conceptualization: Mariana de Moura de Souza and Thiago Bosco Mendes; methodology: Mariana de Moura de Souza and Maria Luiza Rodrigues Defante; formal analysis and investigation: Mariana de Moura de Souza, Rachel Louise Moraes Dantas, and Vitor Leão Durães; writing—original draft preparation: Mariana de Moura de Souza; writing—review and editing: Mariana de Moura de Souza, Rachel Louise Moraes Dantas, Vitor Leão Durães, Maria Luiza Rodrigues Defante, and Thiago Bosco Mendes; supervision: Thiago Bosco Mendes.

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Declarations:

Conflict of Interest: The authors declare no competing interests.

REFERENCES

- Autier P, Boniol M, Pizot C, Mullie P. Vitamin D status and ill health: a systematic review. *Lancet Diabetes Endocrinol*. 2014;2(1):76–89.
- Dawson-Hughes B, Harris SS, Krall EA, Dallal GE. Effect of calcium and vitamin D supplementation on bone density in men and women 65 years of age or older. *N Engl J Med*. 1997;337(10):670–6.
- Holick MF, Binkley NC, Bischoff-Ferrari HA, Gordon CM, Hanley DA, Heaney RP, et al. Evaluation, treatment, and prevention of vitamin D deficiency: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2011;96(7):1911–30.
- LeBoff MS, Greenspan SL, Insogna KL, Lewiecki EM, Saag KG, Singer AJ, et al. The clinician's guide to prevention and treatment of osteoporosis. *Osteoporos Int*. 2022;33(10):2049–102.
- Demay MB, Pittas AG, Bikle DD, Diab DL, Kiely ME, Lazaretti-Castro M, et al. Vitamin D for the Prevention of Disease: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*. 2024;3:dgae290.
- Ross AC, Manson JE, Abrams SA, Aloia JF, Brannon PM, Clinton SK, et al. The 2011 report on dietary reference intakes for calcium and vitamin D from the Institute of Medicine: what clinicians need to know. *J Clin Endocrinol Metab*. 2011;96(1):53–8.
- Bischoff-Ferrari HA, Willett WC, Orav EJ, Lips P, Meunier PJ, Lyons RA, et al. A pooled analysis of vitamin D dose requirements for fracture prevention. *N Engl J Med*. 2012;367(1):40–9.
- Chakhtoura M, Bacha DS, Gharos C, Ajjour S, Assaad M, Jabbour Y, et al. Vitamin D Supplementation and Fractures in Adults: A Systematic Umbrella Review of Meta-Analyses of Controlled Trials. *J Clin Endocrinol Metab*. 2022;107(3):882–98.
- Zhao JG, Zeng XT, Wang J, Liu L. Association Between Calcium or Vitamin D Supplementation and Fracture Incidence in Community-Dwelling Older Adults: A Systematic Review and Meta-analysis. *JAMA*. 2017;318(24):2466–82.
- Hu ZC, Tang Q, Sang CM, Tang L, Li X, Zheng G, et al. Comparison of fracture risk using different supplemental doses of vitamin D, calcium or their combination: a network meta-analysis of randomised controlled trials. *BMJ Open*. 2019;9(10):e024595.
- Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med*. 2009;6(7):e1000097.
- Cumpston M, Li T, Page MJ, Chandler J, Welch VA, Higgins JP, et al. Updated guidance for trusted systematic reviews: a new edition of the Cochrane Handbook for Systematic Reviews of Interventions. *Cochrane Database Syst Rev*. 2019;10(10):ED000142.
- Sanders KM, Stuart AL, Williamson EJ, Simpson JA, Kotowicz MA, Young D. Annual high-dose oral vitamin D and falls and fractures in older women: a randomized controlled trial. *JAMA*. 2010;303(18):1815–1822.
- Waterhouse M, Ebeling PR, McLeod DSA, English D, Romero BD, Baxter C, et al. The effect of monthly vitamin D supplementation on fractures: a tertiary outcome from the population-based, double-blind, randomised, placebo-controlled D-Health trial. *Lancet Diabetes Endocrinol*. 2023;11(5):324–32.
- Joseph P, Pais P, Gao P, Teo K, Xavier D, Lopez-Jaramillo P, et al. Vitamin D supplementation and adverse skeletal and non-skeletal outcomes in individuals at increased cardiovascular risk: Results from the International Polycap Study (TIPS)-3 randomized controlled trial. *Nutr Metab Cardiovasc Dis*. 2023;33(2):434–40.
- Khaw KT, Stewart AW, Waayer D, Lawes CMM, Toop L, Camargo CAJ, et al. Effect of monthly high-dose vitamin D supplementation on falls and non-vertebral fractures: secondary and post-hoc outcomes from the randomised, double-blind, placebo-controlled ViDa trial. *Lancet Diabetes Endocrinol*. 2017;5(6):438–47.
- LeBoff MS, Chou SH, Ratliff KA, Cook NR, Khurana B, Kim E, et al. Supplemental Vitamin D and Incident Fractures in Midlife and Older Adults. *N Engl J Med*. 2022;387(4):299–309.
- Smith H, Anderson F, Raphael H, Maslin P, Crozier S, Cooper C. Effect of annual intramuscular vitamin D on fracture risk in elderly men and women—a population-based, randomized, double-blind, placebo-controlled trial. *Rheumatology (Oxford)*. 2007;46(12):1852–7.
- Trivedi DP, Doll R, Khaw KT. Effect of four monthly oral vitamin D3 (cholecalciferol) supplementation on fractures and mortality in men and women living in the community: randomised double blind controlled trial. *BMJ*. 2003;326(7387):469.
- Luo D, Wan X, Liu J, Tong T. Optimally estimating the sample mean from the sample size, median, mid-range, and/or mid-quartile range. *Stat Methods Med Res*. 2018;27(6):1785–805.
- Le Boff M, Chou S, Murata E, Cook N, Mora S, Lee IM, et al. Effects of vitamin D on the risk of falls in the Vitamin D and Omega-3 Trial (VITAL). *J Bone Miner Res*. 2019;34(Le Boff M.; Chou S.; Murata E.; Cook N.; Mora S.; Lee I.-M.; Buring J.; Manson J.A.) Brigham and Women's Hospital, United States):19.
- Theodoratou E, Tzoulaki I, Zgaga L, Ioannidis JPA. Vitamin D and multiple health outcomes: umbrella review of systematic reviews and meta-analyses of observational studies and randomised trials. *BMJ*. 2014;348:g2035.
- Yao P, Bennett D, Mafham M, Lin X, Chen Z, Armitage J, et al. Vitamin D and Calcium for the Prevention of Fracture: A Systematic Review and Meta-analysis. *JAMA Netw Open*. 2019;2(12):e1917789.

24. **Verlinden L, Carmeliet G.** Integrated View on the Role of Vitamin D Actions on Bone and Growth Plate Homeostasis. *JBMR Plus*. 2021;5(12):e10577.
25. **Christakos S, Dhawan P, Verstuyf A, Verlinden L, Carmeliet G.** Vitamin D: Metabolism, Molecular Mechanism of Action, and Pleiotropic Effects. *Physiol Rev*. 2016;96(1):365–408.
26. **Dent E, Daly RM, Hoogendijk EO, Scott D.** Exercise to Prevent and Manage Frailty and Fragility Fractures. *Curr Osteoporos Rep*. 2023;21(2):205–15.
27. **Wong RMY, Wong H, Zhang N, Chow SKH, Chau WW, Wang J, et al.** The relationship between sarcopenia and fragility fracture—a systematic review. *Osteoporosis International*. 2019;30(3):541–53.
28. **Di Stefano M, Veneto G, Malservisi S, Cecchetti L, Minguzzi L, Strocchi A, et al.** Lactose malabsorption and intolerance and peak bone mass. *Gastroenterology*. 2002;122(7):1793–9.
29. **Katz S, Weinerman S.** Osteoporosis and gastrointestinal disease. *Gastroenterol Hepatol (N Y)*. 2010;6(8):506–17.
30. **Oh HJ, Ryu KH, Park BJ, Yoon BH.** Osteoporosis and Osteoporotic Fractures in Gastrointestinal Disease. *J Bone Metab*. 2018;25(4):213–7.
31. **Gallagher JC.** Effect of early menopause on bone mineral density and fractures. *Menopause*. 2007;14(3 Pt 2):567–71.
32. **Siris ES, Chen YT, Abbott TA, Barrett-Connor E, Miller PD, Wehren LE, et al.** Bone Mineral Density Thresholds for Pharmacological Intervention to Prevent Fractures. *Archives of Internal Medicine*. 2004;164(10):1108–12.
33. **Rothen JP, Rutishauser J, Walter PN, Hersberger KE, Arnet I.** Oral intermittent vitamin D substitution: influence of pharmaceutical form and dosage frequency on medication adherence: a randomized clinical trial. *BMC Pharmacol Toxicol*. 2020;21(1):51.

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